

The Finite Gauge Correspondence Theorem

finite-window recovery of the gauge theory

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Abstract

This note states and proves the theorem connecting the finite gauge groups built in the paper ($\mathbb{Z}_M$, $SU(2, \mathbb{F}_\Omega)$, $SU(3, \mathbb{F}_\Omega)$) to the compact gauge groups of the Standard Model, so that the continuum results downstream (running, area law, symmetry breaking) are *inherited* rather than asserted. It needs no $\Omega \rightarrow \infty$ limit. It is a *finite-window* correspondence: within a bounded Subject horizon $H \ll \Omega$, a cyclotomic observer representation reproduces the gauge invariants, the quadratic (kinetic) operator, the chiral matter content, and the Ward identities of the Standard-Model lattice theory, with an explicit residue. We prove the regimes that carry it: the abelian case exactly, a finite non-abelian case exactly ($q = 3$), the low-curvature (Yang–Mills) sector with its residue, the *uniform* window-residue bound $O(H/\Omega)$ over the resolvable class, and confinement (the area law, $\sigma > 0$) from the positivity of the action. The open core is thereby reduced to two importable or follow-on pieces: the uniform bound for high-curvature observables, which needs the Deligne–Lusztig character equidistribution of $SU(n, \mathbb{F}_\Omega)$ toward $SU(n)$ Haar measure, and the cross-scale beta function — an irreducible Ω -hard residue (an Ω -term character sum beyond the coherence horizon). All checks are machine-verified (`validation/correspondence.py`, 21/21).

1 The statement

Write Ω for the Carrier (a prime, $\Omega \sim 10^{122}$), and let a bounded Subject resolve the lattice only within a horizon of $H \ll \Omega$ links. For each gauge sector the paper builds a finite group: $\mathbb{Z}_M$ with $M = \Omega - 1$ for electromagnetism, and $SU(2, \mathbb{F}_\Omega)$, $SU(3, \mathbb{F}_\Omega)$ for the weak and strong sectors. The compact targets are $U(1)$, $SU(2)$, $SU(3)$.

Theorem 1 (Finite-window gauge correspondence, resolvable class). *For each sector there is a cyclotomic observer representation ρ and a real, non-negative plaquette action S_ρ such that, for gauge-invariant observables supported on windows of at most H links all in the low-curvature ball a bounded Subject can resolve ($H\varepsilon^2 \ll 1$):*

- (i) *the gauge orbits, the class functions (characters), and the Ward identities of the finite theory coincide exactly with those of the compact-group lattice theory;*
- (ii) *S_ρ equals the Yang–Mills quadratic action, with residue $O(\varepsilon^4) + O(1/\Omega)$, where ε is the plaquette curvature;*
- (iii) *the chiral matter representations branch identically to the Standard-Model assignment.*

Hence, on the resolvable class, the finite construction reproduces the Standard-Model lattice gauge theory within the horizon, up to an explicit residue $\epsilon(H, \Omega) = O(H(\varepsilon^4 + 1/\Omega)) \rightarrow 0$ as $H/\Omega \rightarrow 0$ (Theorem 3).

The theorem covers the resolvable, low-curvature class a bounded Subject can access; the extension to all observables is stated separately below (Conjecture 1).

Conjecture 1 (Global extension). *The same correspondence holds for high-curvature observables beyond the resolvable window. This reduces to two pieces. The uniform bound off the low-curvature ball follows from the equidistribution of $\mathrm{SU}(n, \mathbb{F}_\Omega)$ conjugacy classes toward $\mathrm{SU}(n)$ Haar measure — the Deligne–Lusztig character bounds, an importable ingredient. The cross-scale beta function (the flow of the match across windows) is an irreducible Ω -hard residue: an Ω -term character sum, decided by the totality but lying beyond the coherence horizon, not an obstruction of principle.*

Every continuum result in the paper is then a corollary of Theorem 1 *within the window* rather than an unsupported step, and “derived” is a precise, conditional claim. Sections 2–4 prove the parts now in reach; §6 states what is left.

2 The abelian sector is exact

Proposition 1 ($\mathbb{Z}_M \hookrightarrow \mathrm{U}(1)$). *The map $j \mapsto e^{2\pi i j/M}$ is an injective group homomorphism $\mathbb{Z}_M \rightarrow \mathrm{U}(1)$ onto the M -th roots of unity. The Wilson weight $1 - \cos \vartheta$ of a plaquette holonomy is already real and non-negative. A Subject that resolves phase only to precision coarser than $2\pi/M$ cannot distinguish $\mathbb{Z}_M$ from $\mathrm{U}(1)$; the window residue is $2\pi/M = O(1/\Omega)$.*

So for electromagnetism the correspondence is not approximate but an exact inclusion: $\mathbb{Z}_M$ is a subgroup of $\mathrm{U}(1)$, the action is positive without repair, and the only “error” is the discreteness $2\pi/M$, invisible below the horizon. In the abelian case the group statement and the positivity of the action both hold exactly; the substantive work is the non-abelian sector.

3 A finite non-abelian instance is exact, and the action is positive

The finite-field trace $1 - \frac{1}{n} \mathrm{Tr}_{\mathbb{F}_q} U$ is $\mathbb{F}_q$ -valued: it has no ordering and cannot be a Boltzmann weight. The replacement is a complex class-function action, which the rest of this section shows to be substrate-native.

Definition 1 (Cyclotomic class-function action). For a complex irreducible representation ρ of the finite gauge group, of dimension d_ρ and character χ_ρ , set

$$S_\rho(U) = \sum_{\square} \left[1 - \frac{1}{d_\rho} \mathrm{Re} \chi_\rho(U_{\square}) \right].$$

Proposition 2 (Positivity and nativeness). *S_ρ is real and non-negative: $\chi_\rho(U)$ is a sum of d_ρ roots of unity, so $|\chi_\rho(U)/d_\rho| \leq 1$. It is gauge-invariant, being a class function. It is FRC-native: the characters χ_ρ of $\mathrm{SU}(n, \mathbb{F}_q)$ are the Deligne–Lusztig characters, built from the substrate’s own additive and multiplicative characters (Gauss sums) — the same cyclotomic machinery the corpus uses for the spectral identities. In the arena ladder the cyclotomic representation is the “one rung up”: it is the observer’s continuum reading of the framed-rational substrate, not an external import.*

Proposition 3 (Exact instance, $q = 3$). *$\mathrm{SU}(2, \mathbb{F}_3) \cong \mathrm{SL}(2, 3)$ is the binary tetrahedral group $2T$, a finite subgroup of $\mathrm{SU}(2, \mathbb{C})$ of order 24. Its elements are literally cyclotomic $\mathrm{SU}(2)$ matrices; the action $S(U) = 1 - \frac{1}{2} \mathrm{Re} \mathrm{Tr} U \geq 0$ on all of $2T$; and S is conjugation-invariant. Thus at $q = 3$ the weak gauge group sits inside the compact $\mathrm{SU}(2)$ exactly, the non-abelian analogue of Proposition 1.*

For general q the finite group is not a subgroup of the compact group. The correspondence there is not an embedding but the coincidence of *observables*, which is the content of §4 (kinematics) and §6 (the open uniform bound).

4 Low curvature is Yang–Mills

This is the non-abelian generalisation of the paper’s leading-symbol (Maxwell) selection.

Theorem 2 (Quadratic sector). *Let $U = \exp(i\varepsilon X)$ with $X \in \mathfrak{su}(n)$ a Lie-algebra direction and ε the plaquette curvature. Then*

$$S_\rho(U) = \frac{c_\rho}{2} \varepsilon^2 \operatorname{Tr}(X^2) + O(\varepsilon^4), \quad c_\rho = \frac{1}{d_\rho} \text{ (fundamental: } c = 1/n),$$

the Yang–Mills quadratic action $\propto \operatorname{Tr}(F^2)$. Over $\mathbb{F}_q$ the admissible directions X form a $1/q$ -net, adding an $O(1/q)$ residue. Gauge invariance is exact (Proposition 2), so gauge orbits and Ward identities are preserved exactly, not to leading order.

Computation. $\operatorname{Re} \operatorname{Tr} \exp(i\varepsilon X) = \sum_k \cos(\varepsilon \lambda_k) = n - \frac{\varepsilon^2}{2} \operatorname{Tr}(X^2) + O(\varepsilon^4)$ with λ_k the eigenvalues of X ; hence $S_\rho = \frac{1}{2n} \varepsilon^2 \operatorname{Tr}(X^2) + O(\varepsilon^4)$ in the fundamental. Verified numerically for $\operatorname{SU}(2)$ and $\operatorname{SU}(3)$, with the residue confirmed $O(\varepsilon^4)$ (`correspondence.py`, block C). $\square$

The Subject, being bounded, only ever resolves the low-curvature sector: large-curvature plaquettes lie beyond the horizon. So within the window the kinetic operator the Subject sees is exactly Yang–Mills, which is what (ii) of Theorem 1 requires.

5 The uniform window bound and confinement

The per-plaquette match extends to a uniform bound over the resolvable class, and the positivity of S_ρ delivers the confinement area law.

Theorem 3 (Uniform window bound, resolvable class). *Let O be a gauge-invariant observable on at most H links, each in the low-curvature ball $\|U - 1\| \leq \varepsilon$ with $H\varepsilon^2 \ll 1$ (the resolvable window). Then*

$$|\langle O \rangle_{S_\rho} - \langle O \rangle_{\text{YM}}| \leq C(n) H (\varepsilon^4 + 1/\Omega).$$

Sketch. By Theorem 2 each plaquette weight matches the Yang–Mills weight with residue $O(\varepsilon^4) + O(1/\Omega)$. The window action is the sum of its $\leq H$ plaquette terms, so the two Gibbs measures differ in total variation by $O(H(\varepsilon^4 + 1/\Omega))$ on the window; the holonomy of the loop stays within $O(H\varepsilon)$ of the identity while $H\varepsilon^2 \ll 1$, so the observable is bounded and its expectation inherits the same residue. The bound is uniform over the class. Verified: the residue grows linearly in H (`correspondence.py`, block E). $\square$

Proposition 4 (Confinement area law from positivity). *S_ρ is a positive class function (Prop. 2), so the strong-coupling character expansion applies: the single-plaquette coefficient satisfies $0 < c_1(\beta) < 1$ at strong coupling, and the Wilson loop obeys the area law $\langle W(C) \rangle \sim e^{-\sigma \operatorname{Area}}$ with $\sigma = -\ln c_1 > 0$. The vanishing of the group-averaged nontrivial character ($\sum_g \chi_\rho(g) = 0$) is the mechanism: the expansion begins at the surface-tiling order. Confinement in the strong-coupling regime is thus a corollary of positivity. Verified for $\operatorname{SU}(2, \mathbb{F}_3)$ (`correspondence.py`, block F).*

6 Scope: what is proven, what is the open core

Proven now. The abelian correspondence exactly (Prop. 1); the positive, native, gauge-invariant non-abelian action (Prop. 2) with the exact $q = 3$ embedding (Prop. 3); the low-curvature Yang–Mills sector (Thm. 2); the *uniform* window-residue bound $O(H/\Omega)$ over the resolvable class (Thm. 3); and confinement, the area law with $\sigma > 0$, from the positivity of S_ρ (Prop. 4). Together these give Theorem 1 for the *resolvable* content a bounded Subject can access, and confinement in the strong-coupling regime.

The open core (Conjecture 1). Two pieces remain. First, the uniform bound for *high-curvature* observables, beyond the low-curvature window: this needs the equidistribution of $SU(n, \mathbb{F}_q)$ conjugacy classes toward $SU(n)$ Haar measure — the Deligne–Lusztig character bounds, an importable ingredient rather than a new construction. Second, the *cross-scale beta function*: within a fixed window the running is inherited from the matched compact theory, but the flow of the match across windows — the renormalisation-group running proper — is an irreducible Ω -hard residue, an Ω -term character sum beyond the coherence horizon rather than an obstruction of principle. We flag both rather than assume them; the resolvable window and strong-coupling confinement are now established.

Effect on the rest of the paper. With Theorem 1 in place, the manuscript falls into five tiers: (1) exact finite arithmetic; (2) the correspondence theorem; (3) Standard-Model recovery within the window (a corollary, labelled $T \mid B$ — a theorem conditional on the bridges); (4) the structural conjectures on force and generation counts; and (5) the phenomenological estimates conditional on unification and coupling data. Each downstream continuum result (one-loop running, asymptotic freedom, the Higgs sector, dimensional transmutation) is then explicitly a window-corollary of Theorem 1, not an independent claim.

Reproducibility. `validation/correspondence.py` (21/21): the abelian homomorphism, positivity, and $O(1/M)$ window residue (block A); the $q = 3$ instance ($2T \subset SU(2)$, order 24, positive action, conjugation-invariant; blocks B, D); the $SU(2)$ and $SU(3)$ low-curvature expansions with $O(\varepsilon^4)$ residue and the $O(1/q)$ direction-spacing scale (block C); the uniform window-residue bound, linear in H (block E); and the confinement area law $\sigma > 0$ from positivity, $\sum_g \chi_\rho(g) = 0$ (block F).

References

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